

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of professional experience as a Software Developer and Data Engineer. Possesses a Master of Science in Computer Science with expertise in building scalable microservices and distributed data pipelines across various domains. Demonstrates a strong command of Java, Python, Spark, SQL, AWS, and Azure, coupled with experience in CI/CD automation and graduate-level work on large-scale data platforms and privacy-preserving AI applications, with a keen interest in contributing to system design and infrastructure automation for roles like Software Engineer - Registrar.

EDUCATION

Central Michigan University Mount Pleasant, MI Master of Science Computer Science	Aug 2024 - Dec 2025
Lovely Professional University Punjab, India Bachelor of Technology Computer Science and Engineering	Aug 2016 - May 2020

EXPERIENCE

Software Developer NTT DATA Bengaluru, India (Remote)	August 2022 - August 2024
- Designed and optimized high-performance REST and SOAP microservices using Java 8/11 and Spring Boot, handling over 50,000 daily API requests with 99.99 percent SLA uptime, leveraging AWS services for deployment. - Built and automated robust CI/CD pipelines using Jenkins, Maven, and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments, ensuring system stability. - Modernized legacy systems by migrating to Log4j 2.x and improving monitoring and error handling, reducing issue resolution time by 20 percent and enhancing system observability with tools like Kibana. - Led a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles and implementing workflow optimizations. - Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit testing, maintaining high code quality and system integrity.	
Senior Analyst Capgemini Mumbai, India (Remote)	October 2021 - August 2022
- Engineered Java and EJB-based enterprise applications and RESTful web services for automotive clients, contributing to critical system functionalities. - Improved system performance by 90 percent through SQL query optimization on PostgreSQL, database tuning, and modernization of legacy Java code, enhancing user experience. - Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates, significantly reducing deployment risks and improving release velocity. - Mentored junior developers through code reviews, debugging sessions, and clean coding practices, fostering a culture of continuous learning and system design excellence.	
Analyst Capgemini Mumbai, India (Remote)	July 2020 - October 2021
- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization, ensuring continuous service availability. - Enhanced SQL data pipelines and database operations on IBM DB2, improving query performance by 10 percent and strengthening data integrity for critical reporting. - Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage and ensuring application reliability. - Supported release planning and sprint execution using JIRA within Agile development teams, facilitating efficient project workflows and collaboration.	

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Flask, AWS, Terraform)	January 2024 - May 2025
- Built a distributed Spark-based data pipeline to process and analyze more than 8 GB of NYC taxi trip data in near real time, leveraging PySpark for efficient data manipulation. - Developed an interactive Flask web application for data exploration with dynamic filtering and visualization, integrated with Tableau for enhanced analytics. - Containerized ingestion, processing, and analytics components using Docker for reproducible deployments and managed infrastructure with Terraform across AWS services like S3, RDS, ECS, and Lambda. - Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent and improving system scalability. - Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, ensuring robust system design.	
Privacy-Preserving AI Email Client (Capstone Project) (T5, LoRA, Hugging Face)	January 2025 - December 2025
- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, utilizing Hugging Face libraries. - Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, ensuring secure data handling. - Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, demonstrating expertise in privacy-preserving LLM-based applications. - Explored Grafana and Prometheus for monitoring model performance and system health during development and testing phases.	

SKILLS

- Programming Languages: Java, Python, SQL, Go, Ruby, Typescript.
- Frameworks and Libraries: Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit, Flask.
- Web Services: REST API, SOAP API, DNS, DNSSEC.
- Cloud Platforms: AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform, Cloudflare.
- Databases: PostgreSQL, Oracle Database, IBM DB2, RDS, Relational Databases.
- Tools and Technologies: Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, Terraform, Kibana, Grafana, Prometheus.
- Concepts: ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, Privacy-preserving LLM-based applications, System Design, Workflow, Analyzing, Hardware.